

PRODUCT REQUIREMENT DOCUMENT (PRD)

Project: Personal Portfolio Website Integration (Engineering & Data Analytics)

1. DOCUMENT CONTROL & OVERVIEW

Project Name	Anish Lamsal Personal Portfolio Platform	Target Domain	anishlamsal.com.np
Owner	Anish Lamsal	Timeline	Pre-Fall 2026 Deployment
Primary Role	Agricultural Engineer	Future Track	M.S. Data Analytics (Binghamton University)

1.1 Purpose

This document establishes the strategic, visual, functional, and technical specifications for the production-ready personal portfolio website of Anish Lamsal. The site acts as a professional digital identity clearinghouse, bridging full-scale traditional infrastructure engineering achievements with data analytics and visualization capabilities ahead of his graduate program entry at Binghamton University in Fall 2026.

1.2 Target Audience

- **Technical Hiring Managers & Recruiters:** Individuals seeking multi-disciplinary candidates possessing domain analytics expertise paired with production engineering backgrounds.
- **Academic Panels & Advisors:** Research leads evaluating technical profiles for advanced projects or graduate assistantships.

2. VISUAL DESIGN & UI THEME TOKENS

Design parameters are derived directly from theme palettes extracted from premium developer web profiles, balancing high-contrast modern layout structures with clean data scannability.

Visual Token	Value Specification	Functional Scope
Primary Theme Blue	#007acc / #0284c7	Hero elements, button states, link interactions, and visual focus highlights.
Deep Corporate Accent	#0f4c81 / #1a365d	Section banner components, primary headers, block structures.
Canvas Background	#ffffff	

Visual Token	Value Specification	Functional Scope
		Base background level providing premium whitespace presentation.
Muted Canvas Section	#f8fafc	Alternating layout dividers (About Section, Projects section card containers).
Primary Typography Tone	#1e293b	Headers and high-readability body block layouts.
Secondary Typography Tone	#64748b	Subtext, inline geographic location data, metadata badges, timestamps.

2.2 Typography & Interactivity

- **Font System:** Clean sans-serif configuration (`Inter` , `System-UI` , or `Helvetica Neue`).
- **Micro-interactions:** Smooth 200ms transition transforms (e.g., `scale-102` elevates components on cursor grid focus). Includes a native layout theme switcher (Light/Dark canvas) in the navigation framework.

3. STRUCTURAL ARCHITECTURE & CONTENT MODULES

The solution is planned as an elegant Single Page Application (SPA) driven by smooth-scrolling addressable structural link hashes.

3.1 Global Navigation Header

A persistent, sticky layout element anchoring the left text brand indicator ([AnishLamsal](#)) to structural jump anchors ([#home](#) , [#about](#) , [#skills](#) , [#projects](#) , [#experience](#) , [#contact](#)) paired alongside the native theme-switching toggle icon.

3.2 Home/Hero Module

- **Visual Core:** Clean circular border-framed headshot portrait placed with perfect layout balance.
- **Headline Element:** High-weight font reading "Hi, I'm [Anish Lamsal](#)".
- **Dynamic Typewriter Hook:** Interactive subtitle rotating sequentially between:
"I'm an [Agricultural Engineer](#)" → "I'm a [Future Data Analyst](#)" → "I'm a [GIS Specialist](#)".
- **Action Hooks:** Symmetrical CTA arrangement featuring a solid primary button ([\[View My Work\]](#) → [#projects](#)) and a secondary bordered button ([\[Get In Touch\]](#) → [#contact](#)).

3.3 About Me Module

Positioning Narrative Copy:

"Hello! I am Anish Lamsal, an Agricultural Engineer transitioning into the data analytics field as an M.S. Data Analytics student at Binghamton University starting Fall 2026. I blend field engineering expertise with technical data manipulation to optimize infrastructures, map geospatial terrain profiles, and reveal insights hidden within complex datasets. My experience spans leading multi-ton infrastructure design installations for government ministries, supervising mechanized farming deployments, and executing specialized agricultural field research in remote rural areas."

Action Element: Interactive [\[Download CV\]](#) primary button mapping to a secure cloud-hosted copy of the structural PDF resume asset.

3.4 Skills Matrix Grid

A cleanly aligned horizontal-to-vertical structured matrix highlighting technical competencies through specific badge iconography indicators:

- **Core Programming Array:** Python (Pandas, NumPy), Structured Query Language (SQL)
- **Analytical Operations Architecture:** Advanced Data-modeling Excel, ArcGIS Mapping, Tableau Visualization

- **Engineering & Documentation Frameworks:** AutoCAD Design, Technical Reporting, GIS Mapping Systems

4. TECHNICAL PROJECTS SHOWCASE

Structured as a high-impact multi-card layout utilizing distinct color banner headers and modern interface tag lines.

Card Module 1: 1,000-Ton Crop Storage & Infrastructure Layout

- **Header Visual Token:** Solid Dark Theme Accent Blue.
- **Functional Scope:** Supervised the comprehensive structural installation execution of major engineering infrastructures for the Ministry of Industry, Agriculture, and Co-operatives, implementing storage systems saving 18,000 bags of crop assets from post-harvest deterioration.
- **Metadata Categorization Badges:** Infrastructure Design · Project Management · AutoCAD
- **Action Interaction:** [View Technical Documentation] link reference.

Card Module 2: Mini Tiller & Farm Mechanization Research Study

- **Header Visual Token:** Solid Light Theme Blue.
- **Functional Scope:** Research Lead analyzing rural survey configurations across regional municipalities, producing data reports detailing operational bottlenecks to support localized technical development panels.
- **Metadata Categorization Badges:** Field Survey · Data Collection · Technical Reporting
- **Action Interaction:** [View Research Profile] link reference.

Card Module 3: Advanced Agricultural Systems Portfolio

- **Header Visual Token:** Muted Intermediate Blue-Grey.
- **Functional Scope:** Multi-axis engineering training modules completed within intensive laboratory programs hosted at IIT Kharagpur, designing scalable sustainable modern tools.
- **Metadata Categorization Badges:** Sustainable Technologies · Applied Engineering · R&D
- **Action Interaction:** [View Program Validation] link reference.

5. WORK HISTORY & EDUCATIONAL TIMELINE

A unified, interactive vertical layout tracking timeline progression seamlessly across operational engineering milestones and academic tracks.

5.1 Professional Engineering Benchmarks

- **Agricultural Engineer** | Ministry of Industry, Agriculture and Co-operatives, Koshi Province, Nepal
May 2021 – March 2024 (Biratnagar, Nepal)
Led end-to-end design configurations for large storage structures, automated irrigation modules, and automated machine subsidies through extensive on-site structural testing.
- **Engineering Intern** | Bar Pipal Engineering Consultancy
September 2018 – April 2021 (Itahari, Nepal)
Managed road layout field calculations, automated terrain mapping workflows, and optimized stakeholder engineering communications.

5.2 Educational Framework

- **Master of Science in Data Analytics** | Binghamton University, State University of New York
Expected Graduation: December 2028 (Binghamton, NY)
- **Master of Business Administration (MBA)** | KFA Business School, Lincoln University College
Conferred: July 2024 (Kathmandu, Nepal)
- **Bachelor of Agricultural Engineering** | Purwanchal Campus, Tribhuvan University
Conferred: December 2021 (Dharan, Nepal)

6. CORE NON-FUNCTIONAL REQUIREMENTS (NFRS)

- **Performance Parameters:** Target a First Contentful Paint (FCP) velocity threshold < 1.2 seconds. Image optimization pipelines must convert raw imagery into downscaled `.webp` or asset-optimized vector formats.
- **Viewport Layout Fluidity:** Absolute responsiveness mapping across standard mobile views (375px–430px), structural tablets (768px), and high-resolution layout screens (1440px+). Minimum target footprints must scale to $\geq 44 \times 44$ px.
- **Deployment Layer:** Built with clean semantic HTML5, utility CSS styling frameworks (TailwindCSS), and modular JavaScript. Hosted via high-performance static environments (GitHub Pages / Vercel) mapped explicitly to the registered `anishlamsal.com.np` network address.

7. PLATFORM CONTACT LAYOUT

A centered layout block designed to match your visual theme cards, prompting visitors with an inviting closing remark:

"I'm always open to discussing new projects, analytical modeling ideas, or research workflows. Feel free to reach out!"

alamsal@binghamton.edu | [linkedin.com/in/anishlamsal](https://www.linkedin.com/in/anishlamsal)